

THERMAL DECOMPOSITION REACTIONS OF METAL CARBOXYLATO COMPLEXES IN THE SOLID STATE. III. THERMOGRAPHIC AND DIFFERENTIAL THERMAL STUDIES OF METAL OXALATO, MALONATO AND SUCCINATO COMPLEXES

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ABSTRACT

The thermal properties of metal carboxylato complexes of the first transition metals, Mn(II), Fe(II), Fe(III), Co(II), Ni(II) and Cu(II), and non-transition metals, Zn(II) and Cd(II), in the solid state were studied under non-isothermal conditions in a nitrogen atmosphere by simultaneous TG and DTA. TG and DTA curves showed that the decomposition took place in single or multiple steps and thermal stability of the complex decreased, approximately, with the increase of the standard potential of the central metal ion. Thermodynamic parameters, such as activation energy, E_a^* , enthalpy change, ΔH , and entropy change, ΔS , for dehydration and decomposition were determined from TG and DTA curves by standard methods. A linear correlation was found between ΔH and ΔS , and E_a^* and ΔS , in dehydration and decomposition processes. An irreversible phase transition was observed for $\text{Li}_2[\text{Co}(\text{suc})_2]$ in the DTA curve. The residual products were mixtures of Li_2CO_3 and metal oxides except $\text{Li}_2[\text{Mn}(\text{ox})_2]$, $\text{Li}_2[\text{Zn}(\text{mal})_2]$ and $\text{Li}_2[\text{Cd}(\text{mal})_2]$, where the final products were a mixture of $\text{Li}_2\text{O} \cdot \text{MnO}_2$, $\text{Li}_2\text{CO}_3 \cdot \text{ZnCO}_3$ and $\text{Li}_2\text{CO}_3 \cdot \text{CdCO}_3$, respectively.

INTRODUCTION

Thermal investigations of metal oxalato complexes in the solid state under non-isothermal conditions have been carried out by several researchers [1–9]. However, it has been found that studies on metal malonato and succinato complexes are very scarce [8,9]. Complexes prepared were of the type: $\text{Li}_2[\text{ML}_2] \cdot n\text{H}_2\text{O}$ and $\text{Li}_3[\text{FeL}_3] \cdot n\text{H}_2\text{O}$, where M = Mn(II), Fe(II), Co(II), Ni(II), Cu(II), Zn(II) and Cd(II), L = oxalate, malonate and succinate ligands and $n = 1–7$. They were characterized by elemental analyses, IR spectral data, and thermal investigations were made under non-isothermal conditions in a nitrogen atmosphere in the solid state.

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The conversion $[\text{FeL}_3]^{3-} \rightarrow [\text{FeL}_2]^{2-}$ occurred by an electron transfer from a coordinated ligand ion to the central metal ion. Some useful conclusions were made from the thermodynamic parameters such as activation energy, E_a^* , enthalpy change, ΔH , and entropy change, ΔS .

EXPERIMENTAL

The complexes of the type, $\text{Li}_2[\text{ML}_2] \cdot n\text{H}_2\text{O}$ and $\text{Li}_3[\text{FeL}_3] \cdot n\text{H}_2\text{O}$ were synthesized by a general method *. The hydrated metal salt (2.0 mmol) in distilled water (10 cm^3) was mixed with a solution of ligand (40 mmol) in the same solvent (10 cm^3) and the mixture was stirred for $\frac{1}{2}$ h for complete precipitation. The product was collected, washed with water and ethanol and dried in a desiccator (yield ≈ 65 –70%). The chemicals used were obtained from B.D.H. Certain modifications were made for malonato complexes due to their high solubilities in aqueous media. The complexes were characterized by elemental analyses [10] with standard semimicro techniques (Table 1). The thermogravimetric (TG) and differential thermal analysis (DTA) curves were recorded simultaneously using a Shimadzu DT-30 thermal analyser (Japan) in a constant flow of nitrogen with a Pt-crucible using $\alpha\text{-Al}_2\text{O}_3$ as the standard, keeping the heating rate at $10^\circ\text{C min}^{-1}$ for each run, and using 10–20 mg powdered sample.

Activation energies (E_a^*) were calculated from TG and DTA curves using standard methods. Enthalpy changes (ΔH) were calculated from the DTA peak area using indium metal as the calibrant. IR spectra were recorded in KBr media with a Perkin-Elmer 783 IR spectrometer, and carbon was analysed by a Perkin-Elmer 240 C elemental analyser.

RESULTS AND DISCUSSION

TG and DTA studies

TG and DTA curves of some metal oxalato, malonato and succinato complexes are shown in Figs. 1–3. Cd(II) and Zn(II) complexes of malonate and succinate have no lattice water. The remaining complexes in all the series are trihydrates, except $\text{Li}_2[\text{Zn}(\text{mal})_2] \cdot \text{H}_2\text{O}$, $\text{Li}_2[\text{Cu}(\text{ox})_2] \cdot 2\text{H}_2\text{O}$, $\text{Li}_2[\text{Cd}(\text{ox})_2] \cdot 4\text{H}_2\text{O}$, $\text{Li}_2[\text{Fe}(\text{suc})_2] \cdot 4\text{H}_2\text{O}$, $\text{Li}_3[\text{Fe}(\text{ox})_3] \cdot 6\text{H}_2\text{O}$, $\text{Li}_2[\text{Ni}(\text{ox})_2] \cdot 6\text{H}_2\text{O}$, $\text{Li}_2[\text{Cu}(\text{mal})_2] \cdot 6\text{H}_2\text{O}$, $\text{Li}_2[\text{Ni}(\text{suc})_2] \cdot 6\text{H}_2\text{O}$, $\text{Li}_2[\text{Mn}(\text{suc})_2] \cdot 7\text{H}_2\text{O}$ and $\text{Li}_2[\text{Co}(\text{suc})_2] \cdot 7\text{H}_2\text{O}$. All the complexes lost their water molecules before decomposition, except $\text{Li}_2[\text{Cu}(\text{ox})_2] \cdot 2\text{H}_2\text{O}$, where dehydration and decomposition occurred simultaneously in a single

* For Fe(III) complex the ratio of metal salt to ligand was 1:3.

TABLE 1

Analytical data for metal oxalato (ox), malonato (mal) and succinato (suc) complexes

Compound	Elemental analyses (%) ^a	
	Central metal	Carbon
$\text{Li}_2[\text{Mn}(\text{ox})_2] \cdot 3\text{H}_2\text{O}$	18.40 (18.38)	15.98 (16.06)
$\text{Li}_2[\text{Fe}(\text{ox})_2] \cdot 3\text{H}_2\text{O}$	18.60 (18.63)	15.96 (16.01)
$\text{Li}_3[\text{Fe}(\text{ox})_3] \cdot 6\text{H}_2\text{O}$	12.41 (12.44)	16.00 (16.01)
$\text{Li}_2[\text{Co}(\text{ox})_2] \cdot 3\text{H}_2\text{O}$	19.43 (19.46)	15.91 (15.85)
$\text{Li}_2[\text{Ni}(\text{ox})_2] \cdot 6\text{H}_2\text{O}$	16.51 (16.46)	13.42 (13.46)
$\text{Li}_2[\text{Cu}(\text{ox})_2] \cdot 2\text{H}_2\text{O}$	21.92 (21.95)	16.59 (16.58)
$\text{Li}_2[\text{Zn}(\text{ox})_2] \cdot 3\text{H}_2\text{O}$	21.11 (21.13)	15.55 (15.51)
$\text{Li}_2[\text{Cd}(\text{ox})_2] \cdot 4\text{H}_2\text{O}$	30.00 (30.02)	12.78 (12.82)
$\text{Li}_2[\text{Mn}(\text{mal})_2] \cdot 3\text{H}_2\text{O}$	16.78 (16.81)	22.00 (22.02)
$\text{Li}_2[\text{Fe}(\text{mal})_2] \cdot 3\text{H}_2\text{O}$	17.00 (17.04)	21.92 (21.96)
$\text{Li}_3[\text{Fe}(\text{mal})_3] \cdot 3\text{H}_2\text{O}$	12.76 (12.78)	24.68 (24.72)
$\text{Li}_2[\text{Co}(\text{mal})_2] \cdot 3\text{H}_2\text{O}$	17.78 (17.81)	21.80 (21.76)
$\text{Li}_2[\text{Ni}(\text{mal})_2] \cdot 3\text{H}_2\text{O}$	17.71 (17.75)	21.70 (21.77)
$\text{Li}_2[\text{Cu}(\text{mal})_2] \cdot 6\text{H}_2\text{O}$	16.28 (16.31)	18.43 (18.48)
$\text{Li}_2[\text{Zn}(\text{mal})_2] \cdot \text{H}_2\text{O}$	21.61 (21.69)	23.88 (23.89)
$\text{Li}_2[\text{Cd}(\text{mal})_2]$	33.99 (34.02)	21.83 (21.79)
$\text{Li}_2[\text{Mn}(\text{suc})_2] \cdot 7\text{H}_2\text{O}$	12.81 (12.87)	22.40 (22.49)
$\text{Li}_2[\text{Fe}(\text{suc})_2] \cdot 4\text{H}_2\text{O}$	14.92 (14.94)	25.69 (25.68)
$\text{Li}_3[\text{Fe}(\text{suc})_3] \cdot 3\text{H}_2\text{O}$	11.63 (11.66)	30.00 (30.07)
$\text{Li}_2[\text{Co}(\text{suc})_2] \cdot 7\text{H}_2\text{O}$	13.69 (13.67)	22.21 (22.28)
$\text{Li}_2[\text{Ni}(\text{suc})_2] \cdot 6\text{H}_2\text{O}$	14.29 (14.22)	23.30 (23.26)
$\text{Li}_2[\text{Cu}(\text{suc})_2] \cdot 3\text{H}_2\text{O}$	17.52 (17.48)	26.38 (26.41)
$\text{Li}_2[\text{Zn}(\text{suc})_2]$	20.92 (20.99)	30.81 (30.83)
$\text{Li}_2[\text{Cd}(\text{suc})_2] \cdot 3\text{H}_2\text{O}$	27.21 (27.26)	23.21 (23.28)

^a Figures in parentheses are calculated values.

step (Fig. 1, Table 2). TG and DTA curves of all the complexes indicated the single step dehydration except, $\text{Li}_3[\text{Fe}(\text{ox})_3] \cdot 6\text{H}_2\text{O}$, $\text{Li}_2[\text{Cu}(\text{mal})_2] \cdot 6\text{H}_2\text{O}$ and $\text{Li}_2[\text{Cd}(\text{suc})_2] \cdot 3\text{H}_2\text{O}$, where multiple peaks were noticed in DTA curves. The temperature ranges and peak temperatures of the dehydration processes are shown in Tables 2–4. The TG curves of oxalato complexes show a single step decomposition (Table 2), except $\text{Li}_3[\text{Fe}(\text{ox})_3]$ (Fig. 1). A similar phenomenon was also reflected in DTA curves, except $\text{Li}_2[\text{Fe}(\text{ox})_2]$, $\text{Li}_3[\text{Fe}(\text{ox})_3]$ and $\text{Li}_2[\text{Zn}(\text{ox})_2]$, where multiple exothermic peaks were noticed. In malonato and succinato complexes single-step decompositions were followed in TG curves, except $\text{Li}_3[\text{Fe}(\text{mal})_3]$, $\text{Li}_2[\text{Cu}(\text{mal})_2]$, $\text{Li}_2[\text{Cd}(\text{mal})_2]$ and $\text{Li}_2[\text{Cd}(\text{suc})_2]$, where two-stage decompositions were found. The corresponding DTA curves of malonato and succinato complexes showed multiple peaks, except $\text{Li}_2[\text{Co}(\text{mal})_2]$, $\text{Li}_2[\text{Zn}(\text{mal})_2]$ and $\text{Li}_2[\text{Co}(\text{suc})_2]$, where a single peak appeared (Tables 3 and 4). The decomposition of $\text{Li}_2[\text{Cu}(\text{mal})_2]$ was, however, unique, in view of its decomposition through an unstable intermediate, $\text{Li}_2[\text{Cu}_2(\text{mal})_2]$, where copper was stated to be univalent [11].

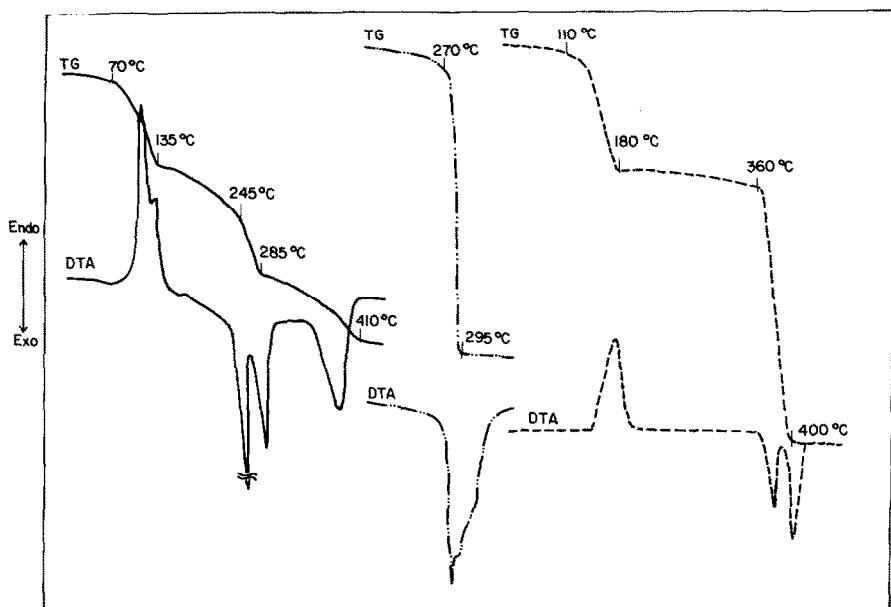


Fig. 1. Thermal curves of $\text{Li}_3[\text{Fe}(\text{ox})_3] \cdot 6\text{H}_2\text{O}$, sample mass 9.85 mg (—), $\text{Li}_2[\text{Cu}(\text{ox})_2] \cdot 2\text{H}_2\text{O}$, sample mass 17.00 mg (· · · · ·) and $\text{Li}_2[\text{Zn}(\text{ox})_2] \cdot 3\text{H}_2\text{O}$, sample mass 18.30 mg (-----).

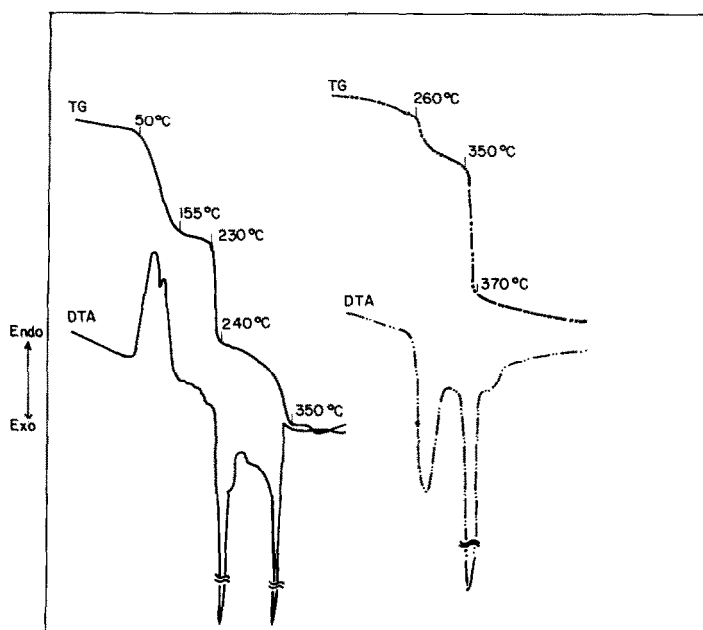


Fig. 2. Thermal curves of $\text{Li}_2[\text{Cu}(\text{mal})_2] \cdot 6\text{H}_2\text{O}$, sample mass 11.53 mg (—) and $\text{Li}_2[\text{Cd}(\text{mal})_2]$, sample mass 14.80 mg (· · · · ·).

TABLE 2

Dehydration and decomposition reactions of metal oxalato complexes

No.	Decomposition reaction	Temperature range (°C)	DTA peak temperature (°C)		E_a^* (kJ mol ⁻¹)	ΔH (kJ mol ⁻¹)	ΔS (J K ⁻¹ mol ⁻¹)
			Endo.	Exo.			
1	$\text{Li}_2[\text{Mn}(\text{C}_2\text{O}_4)_2] \cdot 3\text{H}_2\text{O} \rightarrow \text{Li}_2[\text{Mn}(\text{C}_2\text{O}_4)_2]$	140–210	190		86.8	82.7	97.4
					170.1	130.4	175.7
2	$\text{Li}_2[\text{Fe}(\text{C}_2\text{O}_4)_2] \cdot 3\text{H}_2\text{O} \rightarrow \text{Li}_2[\text{Fe}(\text{C}_2\text{O}_4)_2]$	150–220	200		120.3	129.0	271.7
					225, 255	–	–
3	$\text{Li}_3[\text{Fe}(\text{C}_2\text{O}_4)_3] \cdot 6\text{H}_2\text{O} \rightarrow \text{Li}_3[\text{Fe}(\text{C}_2\text{O}_4)_3]$	70–135	110, 130		63.0	–	197.8 ^a
					38.2	–	–
					219.4	–	–
					380	109.9	74.0
4	$\text{Li}_2[\text{Co}(\text{C}_2\text{O}_4)_2] \cdot 3\text{H}_2\text{O} \rightarrow \text{Li}_2[\text{Co}(\text{C}_2\text{O}_4)_2]$	150–200	190		143.2	122.6	50.0
					246.0	147.4	91.4
5	$\text{Li}_2[\text{Ni}(\text{C}_2\text{O}_4)_2] \cdot 6\text{H}_2\text{O} \rightarrow \text{Li}_2[\text{Ni}(\text{C}_2\text{O}_4)_2]$	80–160	125		66.7	61.6	94.4
					338.2	220.0	68.4
6	$\text{Li}_2[\text{Cu}(\text{C}_2\text{O}_4)_2] \cdot 2\text{H}_2\text{O} \rightarrow \text{Li}_2\text{CO}_3 \cdot \text{CuO}$	270–295		280, 290	–	–	54.2 ^a
					88.7	116.0	46.5
7	$\text{Li}_2[\text{Zn}(\text{C}_2\text{O}_4)_2] \cdot 3\text{H}_2\text{O} \rightarrow \text{Li}_2[\text{Zn}(\text{C}_2\text{O}_4)_2]$	110–180	175		486.7	–	24.5 ^a
					375, 395	–	–
8	$\text{Li}_2[\text{Cd}(\text{C}_2\text{O}_4)_2] \cdot 4\text{H}_2\text{O} \rightarrow \text{Li}_2[\text{Cd}(\text{C}_2\text{O}_4)_2]$	70–135	115		79.0	105.4	62.1
					340	240.0	42.5
	$\text{Li}_2[\text{Cd}(\text{C}_2\text{O}_4)_2] \rightarrow \text{Li}_2\text{CO}_3 \cdot \text{CdO}$	295–355			335.2	240.0	71.0

^a Overall enthalpy change.

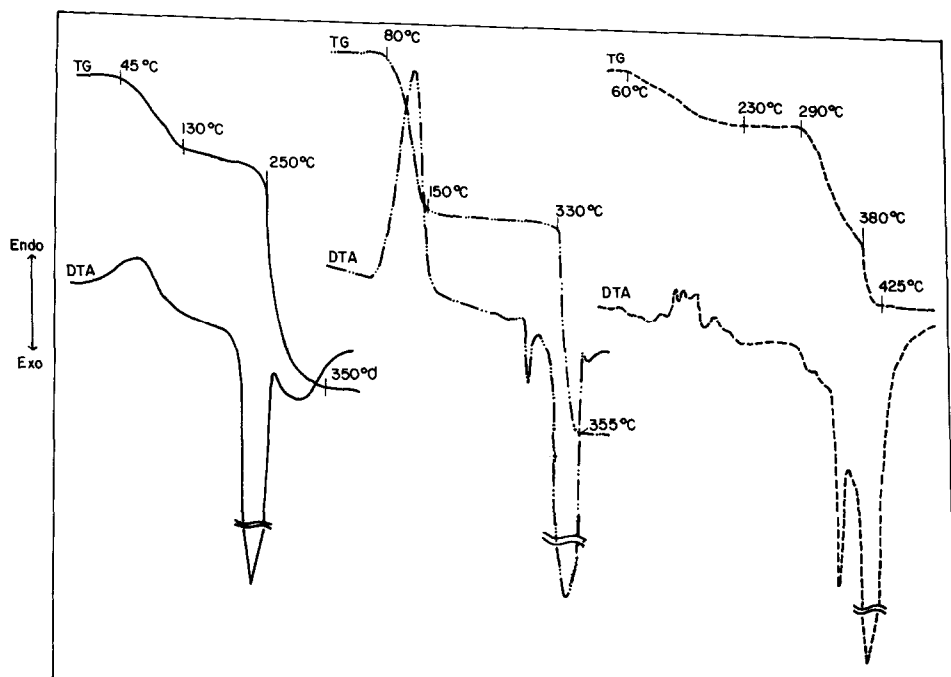
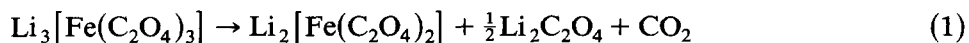


Fig. 3. Thermal curves of $\text{Li}_3[\text{Fe}(\text{suc})_3] \cdot 3\text{H}_2\text{O}$, sample mass 11.50 mg (—), $\text{Li}_2[\text{Co}(\text{suc})_2] \cdot 7\text{H}_2\text{O}$, sample mass 14.80 mg (· · · · ·) and $\text{Li}_2[\text{Cd}(\text{suc})_2] \cdot 3\text{H}_2\text{O}$, sample mass 10.75 mg (-----).

For the DTA curve of $\text{Li}_2[\text{Co}(\text{suc})_2]$ a prominent phase transition was observed (exothermic in nature) showing no weight loss along the TG curve (280–300 °C). Such a phase change in the DTA curve indicates that some sort of structural rearrangement is occurring before the decomposition of the complex.

The transformation of the tris- to the bis-oxalato complex suggested the reaction to be:



The liberation of CO_2 indicated that the central metal ion captured an electron from the ligand ion [2]. A similar sequence was also found in the malonato complex, and the respective DTA peak temperatures were 225 and 230 °C. For $\text{Li}_3[\text{Fe}(\text{suc})_3]$, the intermediate reduction step was not obtained due to its direct conversion to $\text{Li}_2\text{CO}_3 \cdot \text{FeO}$.

For oxalato, malonato and succinato complexes the dehydration peak temperature in either a single step or in the first step of multiple step processes are in the order: $\text{Fe(II)} > \text{Mn(II)} \approx \text{Co(II)} > \text{Zn(II)} > \text{Ni(II)} > \text{Cd(II)} > \text{Fe(III)}$, $\text{Ni(II)} > \text{Co(II)} > \text{Mn(II)} \approx \text{Fe(II)} > \text{Cu(II)} > \text{Zn(II)} > \text{Fe(III)}$ and $\text{Ni(II)} > \text{Co(II)} > \text{Fe(II)} > \text{Mn(II)} > \text{Cd(II)} > \text{Fe(III)} \approx \text{Cu(II)}$, respectively (Tables 2–4) and their respective decomposition peak tempera-

TABLE 3

Dehydration and decomposition reactions of metal malonato complexes

No.	Decomposition reaction	Temperature range (°C)	DTA peak temperature E_a^* (kJ mol ⁻¹)		ΔH (kJ mol ⁻¹)	ΔS (J K ⁻¹ mol ⁻¹)
			Endo	Exo.		
1	$\text{Li}_2[\text{Mn}(\text{C}_3\text{H}_2\text{O}_4)_2] \cdot 3\text{H}_2\text{O} \rightarrow \text{Li}_2[\text{Mn}(\text{C}_3\text{H}_2\text{O}_4)_2]$	150–210	185	128.2	149.2	560.8
	$\text{Li}_2[\text{Mn}(\text{C}_3\text{H}_2\text{O}_4)_2] \rightarrow \text{Li}_2\text{CO}_3 \cdot \text{MnO}_2$	280–360		196.9	—	—
2	$\text{Li}_2[\text{Fe}(\text{C}_3\text{H}_2\text{O}_4)_2] \cdot 3\text{H}_2\text{O} \rightarrow \text{Li}_2[\text{Fe}(\text{C}_3\text{H}_2\text{O}_4)_2]$	60–210	185	55.9	64.4	130.3
	$\text{Li}_2[\text{Fe}(\text{C}_3\text{H}_2\text{O}_4)_2] \rightarrow \text{Li}_2\text{CO}_3 \cdot \text{FeO}$	210–310		301.1	—	—
3	$\text{Li}_3[\text{Fe}(\text{C}_3\text{H}_2\text{O}_4)_3] \cdot 3\text{H}_2\text{O} \rightarrow \text{Li}_3[\text{Fe}(\text{C}_3\text{H}_2\text{O}_4)_3]$	30–170	80	35.2	39.6	206.3
	$\text{Li}_3[\text{Fe}(\text{C}_3\text{H}_2\text{O}_4)_3] \rightarrow \text{Li}_2[\text{Fe}(\text{C}_3\text{H}_2\text{O}_4)_2]$	170–230		70.9	—	—
	$\text{Li}_2[\text{Fe}(\text{C}_3\text{H}_2\text{O}_4)_2] \rightarrow \text{Li}_2\text{CO}_3 \cdot \text{FeO}$	230–340		76.4	—	—
4	$\text{Li}_2[\text{Co}(\text{C}_3\text{H}_2\text{O}_4)_2] \cdot 3\text{H}_2\text{O} \rightarrow \text{Li}_2[\text{Co}(\text{C}_3\text{H}_2\text{O}_4)_2]$	150–210	200	133.4	127.3	298.7
	$\text{Li}_2[\text{Co}(\text{C}_3\text{H}_2\text{O}_4)_2] \rightarrow \text{Li}_2\text{CO}_3 \cdot \text{CoO}$	280–360		241.9	144.9	399.8
5	$\text{Li}_2[\text{Ni}(\text{C}_3\text{H}_2\text{O}_4)_2] \cdot 3\text{H}_2\text{O} \rightarrow \text{Li}_2[\text{Ni}(\text{C}_3\text{H}_2\text{O}_4)_2]$	200–250	240	152.4	173.2	235.3
	$\text{Li}_2[\text{Ni}(\text{C}_3\text{H}_2\text{O}_4)_2] \rightarrow \text{Li}_2\text{CO}_3 \cdot \text{NiO}$	310–347		392.0	—	—
6	$\text{Li}_2[\text{Cu}(\text{C}_3\text{H}_2\text{O}_4)_2] \cdot 6\text{H}_2\text{O} \rightarrow \text{Li}_2[\text{Cu}(\text{C}_3\text{H}_2\text{O}_4)_2]$	50–155	140, 150	51.1	—	—
	$\text{Li}_2[\text{Cu}(\text{C}_3\text{H}_2\text{O}_4)_2] \rightarrow \text{Li}_2[\text{Cu}_2(\text{C}_3\text{H}_2\text{O}_4)_2]$	230–240		—	—	—
	$\text{Li}_2[\text{Cu}_2(\text{C}_3\text{H}_2\text{O}_4)_2] \rightarrow \text{Li}_2\text{CO}_3 \cdot \text{CuO}$	240–350		79.5	—	—
7	$\text{Li}_2[\text{Zn}(\text{C}_3\text{H}_2\text{O}_4)_2] \cdot \text{H}_2\text{O} \rightarrow \text{Li}_2[\text{Zn}(\text{C}_3\text{H}_2\text{O}_4)_2]$	115–155	130	72.5	94.3	69.8
	$\text{Li}_2[\text{Zn}(\text{C}_3\text{H}_2\text{O}_4)_2] \rightarrow \text{Li}_2\text{CO}_3 \cdot \text{ZnCO}_3$	310–350	320	401.8	—	—
8	$\text{Li}_2[\text{Cd}(\text{C}_3\text{H}_2\text{O}_4)_2] \rightarrow \text{Li}_2[\text{Cd}(\text{C}_3\text{H}_2\text{O}_4)_2]$	260–350	300	33.6	46.4	159.2
	$\text{Li}_2[\text{Cd}(\text{C}_3\text{H}_2\text{O}_4)_2] \rightarrow \text{Li}_2\text{CO}_3 \cdot \text{CdCO}_3$	350–370		—	—	—

^a Overall enthalpy change.

TABLE 4

Dehydration and decomposition reactions of metal succinato complexes

No.	Decomposition reaction	Temperature range (°C)	DTA peak temperature (°C)	DTA		E_a^* (kJ mol ⁻¹)	ΔH (kJ mol ⁻¹)	ΔS (J K ⁻¹ mol ⁻¹)
				Endo.	Exo.			
1	$\text{Li}_2[\text{Mn}(\text{C}_4\text{H}_4\text{O}_4)_2] \cdot 7\text{H}_2\text{O} \rightarrow \text{Li}_2[\text{Mn}(\text{C}_4\text{H}_4\text{O}_4)_2]$	35–140	105			84.8	169.5	448.3
	$\text{Li}_2[\text{Mn}(\text{C}_4\text{H}_4\text{O}_4)_2] \rightarrow \text{Li}_2\text{CO}_3 \cdot \text{MnO}_2$	330–370			320, 365	121.7	—	—
2	$\text{Li}_2[\text{Fe}(\text{C}_4\text{H}_4\text{O}_4)_2] \cdot 4\text{H}_2\text{O} \rightarrow \text{Li}_2[\text{Fe}(\text{C}_4\text{H}_4\text{O}_4)_2]$	30–140	110			44.5	98.4	256.8
	$\text{Li}_2[\text{Fe}(\text{C}_4\text{H}_4\text{O}_4)_2] \rightarrow \text{Li}_2\text{CO}_3 \cdot \text{FeO}$	270–470			285, 295	32.6	370.0 ^a	—
3	$\text{Li}_3[\text{Fe}(\text{C}_4\text{H}_4\text{O}_4)_3] \cdot 3\text{H}_2\text{O} \rightarrow \text{Li}_3[\text{Fe}(\text{C}_4\text{H}_4\text{O}_4)_3]$	45–130	90			38.8	162.7	454.4
	$\text{Li}_3[\text{Fe}(\text{C}_4\text{H}_4\text{O}_4)_3] \rightarrow \text{Li}_2\text{CO}_3 \cdot \text{FeO}$	250–350			285, 330	104.0	—	—
4	$\text{Li}_2[\text{Co}(\text{C}_4\text{H}_4\text{O}_4)_2] \cdot 7\text{H}_2\text{O} \rightarrow \text{Li}_2[\text{Co}(\text{C}_4\text{H}_4\text{O}_4)_2]$	80–150	120			69.8	273.9	696.9
	$\text{Li}_2[\text{Co}(\text{C}_4\text{H}_4\text{O}_4)_2] \xrightarrow{\text{Phase change}} \text{Li}_2[\text{Co}(\text{C}_4\text{H}_4\text{O}_4)_2]$	280–300			290	—	20.8	—
	$\text{Li}_2[\text{Co}(\text{C}_4\text{H}_4\text{O}_4)_2] \rightarrow \text{Li}_2\text{CO}_3 \cdot \text{CoO}$	330–355			345	219.2	—	—
5	$\text{Li}_2[\text{Ni}(\text{C}_4\text{H}_4\text{O}_4)_2] \cdot 6\text{H}_2\text{O} \rightarrow \text{Li}_2[\text{Ni}(\text{C}_4\text{H}_4\text{O}_4)_2]$	100–165	140			95.3	469.2	1136.2
	$\text{Li}_2[\text{Ni}(\text{C}_4\text{H}_4\text{O}_4)_2] \rightarrow \text{Li}_2\text{CO}_3 \cdot \text{NiO}$	370–400			375, 385, 395	486.2	757.8 ^a	—
6	$\text{Li}_2[\text{Cu}(\text{C}_4\text{H}_4\text{O}_4)_2] \cdot 3\text{H}_2\text{O} \rightarrow \text{Li}_2[\text{Cu}(\text{C}_4\text{H}_4\text{O}_4)_2]$	45–100	90			68.6	96.3	265.3
	$\text{Li}_2[\text{Cu}(\text{C}_4\text{H}_4\text{O}_4)_2] \rightarrow \text{Li}_2\text{CO}_3 \cdot \text{CuO}$	265–330			320, 325	364.8	333.2 ^a	—
7	$\text{Li}_2[\text{Zn}(\text{C}_4\text{H}_4\text{O}_4)_2] \rightarrow \text{Li}_2\text{CO}_3 \cdot \text{ZnO}$	400–515			440, 490	243.5	689.7 ^a	—
8	$\text{Li}_2[\text{Cd}(\text{C}_4\text{H}_4\text{O}_4)_2] \cdot 3\text{H}_2\text{O} \rightarrow \text{Li}_2[\text{Cd}(\text{C}_4\text{H}_4\text{O}_4)_2]$	60–230	100, 120, 135, 160, 185, 210			24.6	—	—
	$\text{Li}_2[\text{Cd}(\text{C}_4\text{H}_4\text{O}_4)_2] \rightarrow \text{Li}_2[\text{Cd}(\text{C}_4\text{H}_4\text{O}_4)]$	290–380			310, 335, 355	110.7	—	—
	$\text{Li}_2[\text{Cd}(\text{C}_4\text{H}_4\text{O}_4)] \rightarrow \text{Li}_2\text{CO}_3 \cdot \text{CdO}$	380–425			395, 420	477.6	—	—

^a Overall enthalpy change.

tures * follow the trend: $\text{Zn(II)} > \text{Ni(II)} > \text{Cd(II)} > \text{Mn(II)} > \text{Co(II)} > \text{Fe(III)} > \text{Fe(II)}$, $\text{Co(II)} \approx \text{Zn(II)} > \text{Ni(II)} > \text{Cd(II)} > \text{Mn(II)} > \text{Fe(II)} > \text{Fe(III)} \approx \text{Cu(II)}$ and $\text{Zn(II)} > \text{Ni(II)} > \text{Co(II)} > \text{Mn(II)} \approx \text{Cu(II)} > \text{Cd(II)} > \text{Fe(II)} \approx \text{Fe(III)}$.

Activation energies (E_a^*) for dehydration and decomposition were computed from TG curves by Horowitz and Metzger's method [12] and from DTA curves by Borchardt and Daniels' equation [13]. The values are given in Tables 2–4. In most cases values are fairly close to each other, except $\text{Li}_2[\text{Ni(mal)}_2] \cdot 3\text{H}_2\text{O}$, $\text{Li}_2[\text{Zn(ox)}_2] \cdot 3\text{H}_2\text{O}$ and $\text{Li}_2[\text{Fe(suc)}_2] \cdot 4\text{H}_2\text{O}$. Activation energies (E_a^*) for dehydration from DTA curves of oxalato, malonato and succinato complexes follow the sequence: $\text{Fe(II)} > \text{Co(II)} > \text{Zn(II)} > \text{Cd(II)} > \text{Mn(II)} > \text{Ni(II)}$, $\text{Ni(II)} > \text{Mn(II)} > \text{Co(II)} > \text{Zn(II)} > \text{Fe(II)} > \text{Fe(III)}$ and $\text{Fe(II)} > \text{Ni(II)} > \text{Co(II)} > \text{Mn(II)} \approx \text{Cu(II)} > \text{Fe(III)}$, respectively. For decomposition, activation energies (E_a^*) from TG ** curves of oxalato, malonato and succinato complexes follow the order: $\text{Zn(II)} > \text{Ni(II)} > \text{Cd(II)} > \text{Co(II)} > \text{Mn(II)} > \text{Fe(III)}$, $\text{Zn(II)} > \text{Ni(II)} > \text{Fe(II)} > \text{Co(II)} > \text{Mn(II)} > \text{Fe(III)}$ and $\text{Ni(II)} > \text{Cu(II)} > \text{Zn(II)} > \text{Co(II)} > \text{Mn(II)} > \text{Cd(II)} > \text{Fe(III)} > \text{Fe(II)}$, respectively.

For dehydration, the enthalpy changes (ΔH) and entropy changes (ΔS) of oxalato and malonato complexes were in the order: $\text{Fe(III)} > \text{Fe(II)} > \text{Ni(II)} > \text{Cd(II)} > \text{Cu(II)}^{***} > \text{Co(II)} > \text{Zn(II)} > \text{Mn(II)}$ and $\text{Mn(II)} > \text{Cu(II)} > \text{Co(II)} > \text{Ni(II)} > \text{Cd(II)} > \text{Fe(III)} > \text{Fe(II)} > \text{Zn(II)}$, except the position of Cd(II) and Fe(III) is reversed for entropy changes (ΔS) for the malonato series. In the succinato series, the respective order of enthalpy changes (ΔH) and entropy changes (ΔS) are as follows: $\text{Ni(II)} > \text{Co(II)} > \text{Mn(II)} > \text{Fe(III)} > \text{Fe(II)} > \text{Cu(II)}$ and $\text{Ni(II)} > \text{Co(II)} > \text{Fe(III)} > \text{Mn(II)} > \text{Cu(II)} > \text{Fe(II)}$.

Activation energies (evaluated from TG curves [†]) for decomposition of the oxalato and malonato series follow the order: $\text{Zn(II)} > \text{Ni(II)} > \text{Cd(II)} > \text{Fe(II)} > \text{Co(II)} > \text{Mn(II)} > \text{Fe(III)}$, and for the succinato series the order is $\text{Ni(II)} > \text{Cu(II)} > \text{Zn(II)} > \text{Co(II)} > \text{Mn(II)} > \text{Cd(II)} > \text{Fe(III)} > \text{Fe(II)}$.

The enthalpy changes (ΔH) and entropy changes (ΔS) for decomposition processes of oxalato complexes follow the order: $\text{Mn(II)} > \text{Co(II)} > \text{Ni(II)} > \text{Cd(II)} > \text{Zn(II)}$. The overall enthalpy changes for malonato and succinato complexes follow the trend: $\text{Ni(II)} > \text{Fe(II)} > \text{Mn(II)} > \text{Co(II)}$ and $\text{Ni(II)} > \text{Zn(II)} > \text{Fe(II)} > \text{Cu(II)}$, respectively. The order of entropy changes (ΔS) was not evaluated for the multiple peaks in DTA curves.

* First peak is considered for multiple steps of decomposition.

** As DTA values were not available.

*** Overall enthalpy change.

[†] E_a^* from TG curves were considered as very few values from DTA curves were available.

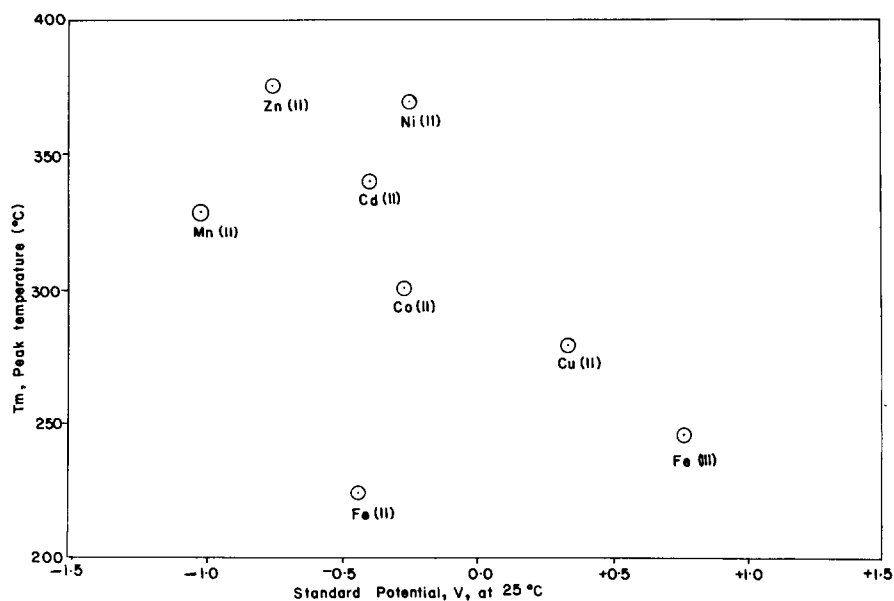


Fig. 4. Plots of peak temperatures (T_m , °C) of decomposition for metal oxalato complexes versus standard potentials, V , of central metal ion at 25°C.

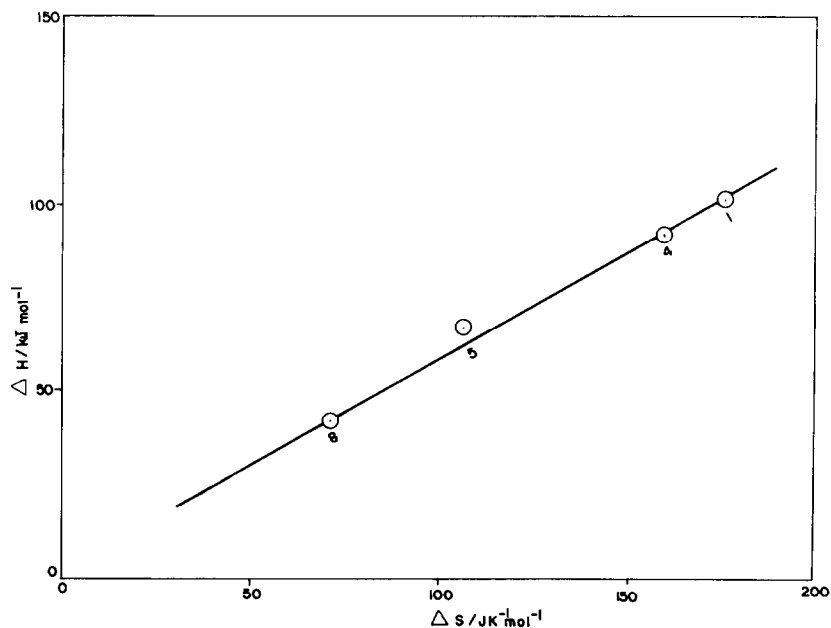


Fig. 5. Plots of ΔH versus ΔS for the thermal decomposition of metal oxalato complexes, 1 = Mn(II), 4 = Co(II), 5 = Ni(II) and 8 = Cd(II).

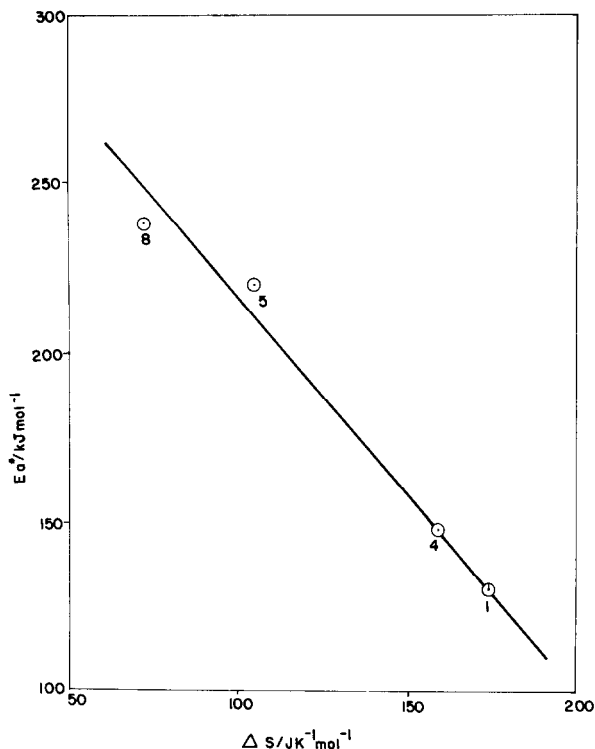


Fig. 6. Plots of E_a^* versus ΔS for the thermal decomposition of metal oxalato complexes, 1 = Mn(II), 4 = Co(II), 5 = Ni(II) and 8 = Cd(II).

By plotting DTA peak temperature (T_m , °C) for decomposition of oxalato complexes against the standard potential of the central metal ions * (Fig. 4), it was found that the thermal stability of the complex decreases, approximately, with the increase in the tendency of the central metal ion to capture an electron from the ligand.

A linear correlation was found by plotting ΔH versus ΔS , as shown in Fig. 5. The values of entropy changes (ΔS) were calculated by the relation [15]:

$$\Delta S = \Delta H / T_m \quad (2)$$

where T_m is the peak temperature of decomposition in the DTA curves. Similarly, a linear relationship was also observed by plotting E_a^* versus ΔS ,

* The plots of DTA peak temperatures of the complexes versus the standard potential [14] of the central metal ion forming an aquo complex are given in Fig. 4. These potentials were considered in order to measure the tendency of the central metal ion to capture an electron from the ligand forming an oxalato complex, since the Dq value of H_2O is near to that of $C_2O_4^{2-}$.

as shown in Fig. 6. It has been observed that a system having higher entropy change (ΔS) will require less activation energy ($E_a^\star$) for its decomposition. A more or less similar sequence was also observed for dehydration processes.

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